



DEPARTMENT OF THE NAVY
NAVAL SURFACE WARFARE CENTER
PHILADELPHIA DIVISION
5001 SOUTH BROAD STREET
PHILADELPHIA PA 19112-1403

IN REPLY REFER TO:

J&A No. 0222-24-160-00

**JUSTIFICATION AND APPROVAL
FOR USE OF OTHER THAN FULL AND OPEN COMPETITION**

1. Contracting Activity.

The Naval Sea Systems Command, Naval Surface Warfare Center, Philadelphia Division (NSWCPD), Code 0222.

2. Description of the Action Being Approved.

Award of a contract and subsequent orders on a sole source basis to Fairmount Automation, Inc., 10 Clipper Road, West Conshohocken, PA 19428 – CAGE Code 09TT2, for the procurement of Original Equipment Manufacturer (OEM) control systems hardware components for the systems listed in Table 1 below. This award is a follow-on requirement to Indefinite-Delivery-Indefinite-Quantity (IDIQ) contract N64498-19-D-4023 awarded to Fairmount Automation for material procurements for multiple ship class Ship Control Systems by the NSWCPD.

Ship Class	Systems
CVN 68	Catapult Steam Charging Control System (CSCCS)
CVN 68	Reboiler Control System (RCS)
CVN 68, LHD 1, AS 39, LCC 19	Auxiliary Steam Control Systems (ASCS)
CVN 68, LHD 1, AS 39, LCC 19	Shaft Speed Indication System (SSIS)
CVN 68, LHD 1, AS 39, LCC 19	Hotwell Level Control Systems (HWLCS)
LHD 1, AS 39, LCC 19, FMS	Automatic Boiler Control (ABC) and Burner Management Control Systems (BMCS)

Table 1: List of Systems

3. Description of Supplies/Services.

The proposed acquisition will provide for procurement of Fairmount Automation model FAC-2100+, and Chameleon+ programmable automation controller (PAC) modules, and accessories. The proposed acquisition will also include a quantity of Windows-based configuration software package licenses from Fairmount Automation. The Windows-based configuration software package licenses are required for various CVN, LHD, LCC, DDG, and AS class ships to communicate with the controllers. Fairmount Automation shall also provide evaluation and repair services in support of any Navy control systems employing Fairmount Automation products as

directed by NSWCPD. A complete list of parts, services and quantities to be procured under this procurement will be attached to this J&A for reference.

The required delivery date for the first order will be in FY25. The anticipated completion date is five (5) years from the date of the award. The requirement will be funded using Operations Maintenance Navy (OM&N) and Ship Conversion Navy (SCN) funds. Appropriate funding will be applied at the delivery order level. The Government's minimum needs have been verified by certifying technical and requirements personnel. The total estimated dollar value of the proposed procurement is [REDACTED] and will be funded as shown on Table 2.

[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]

Table 2: Proposed Contract Funding Plan

4. Statutory Authority Permitting Other Than Full and Open Competition.

10 U.S.C. 3204(a)(1), Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

5. Rationale Justifying Use of Cited Statutory Authority.

In accordance with FAR 6.302-1(a)(2)(ii)(A) and (B), a sole source basis to Fairmount Automation is justified as this requirement is for highly specialized supplies and services when it is likely that award to any other source would result in substantial duplication of cost to the Government that is not expected to be recovered through competition and it would result in unacceptable delays in fulfilling the agency's requirements as detailed below.

Fairmount Automation is the OEM for PAC hardware components currently in use on various ship control systems listed in Table 1 and is the only source that can supply compatible items. The Fairmount Automation systems listed in Table 1 have been installed over the past 25 years to replace older obsolete systems, and this equipment has proven to be reliable and maintainable.

Due to obsolescence issues with electronic components in their hardware, Fairmount Automation needed to develop a next generation control hardware product line. The next generation components are identified with a + after the model number. The + model components are designed to be one for-one-compatible replaceable components that can utilize all the existing tested software programs and system designs created when the current systems were installed. The + versions also are designed to interoperate with the legacy components (models without + designation) on existing systems. Due to the current significant installed base of Fairmount Automation equipment in use on Navy ships, updating with Fairmount equipment is the only reasonable solution for dealing with the obsolescence issue on these Navy control systems. Fairmount Automation equipment is the only company that can supply equipment that will operate and interface on the Navy control systems already in operation on the listed ship classes without significant modification. This was confirmed through market research. Any other equipment would require a complete redesign of the system that would take at least a decade and result in exorbitant costs.

This new contract is a follow-on requirement to Contract No. N64498-19-D-4023 awarded on a sole source basis to Fairmount Automations on 26 June 2019 under the authority of J&A No. 0221-18-233-00 approved by Karen Gutmaker. That contract was the third contract over the past 20 years awarded to Fairmount Automation, for material procurements for multiple ship class control systems by the Naval Surface Warfare Center Philadelphia Division. Under prior contracts, Fairmount Automation provided proprietary hardware for the systems listed in Table 1, for which it was the primary designer and producer. This new sole source contract meets both of the FAR Subpart 6.302-1(a)(2)(ii) (A) and (B) conditions as discussed in the following paragraphs.

Due to the highly specialized unique design of the Fairmount Automation products, no alternate supplier has equipment that can be used as a direct one-for-one replacement. The government would need to completely redesign these systems for the Navy to replace the Fairmount Automation equipment with an alternate supplier's equipment. The engineering required to redesign, develop, and test each new Navy critical system is a significant effort that would cost many times more than the Fairmount Automation hardware costs. Also, the effort to remove the Fairmount Automation systems and install an alternate supplier's replacement system would also cost many times more than the Fairmount Automation hardware costs. There are also other associated logistics development and training costs that would be incurred if an alternate supplier's equipment were to be used. Regarding substantial duplication of cost, the below table shows the cost estimates associated with designing and installing an alternate design to include: alternate supplier's material costs, engineering design and development, shipboard removal of Fairmount Automation systems and installation of replacement systems, shipboard testing and certification, and logistical support through an alternative source. The total cost associated with using an alternate supplier's equipment for all ship classes and unique systems listed in Table 3 is estimated to be [REDACTED]. This cost is not expected to be recovered through competition.

System	Alternate Supplier Material	Engineering Design and Development	Removal and Installation	Shipboard Testing	Logistics and Training	Totals
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]

Table 3: Estimated Alternate Vendor System Development (Cost Impact)

There is at least ten (10) years needed to complete the planning and estimating, engineering and logistics development, and installation and testing times needed to replace all the existing Fairmount Automation systems noted in Table 1. Table 4 lists the approximate times needed for planning and estimating, engineering and logistics development, and shipboard installation and testing. Planning and estimating involves drafting detailed cost proposal development, submittal,

and funding request presentations. Due to the Fleet budgeting process (request and approval of funding) for projects of this magnitude, the costs would need to be spread over several years and would take over one (1) year from project proposal approvals until the first increment of funding is received. If started immediately, the planning and estimating time and the one-year delay before receipt of funding would delay the engineering effort until 2026 at the earliest. The engineering and logistics updates could start in 2026, but they would not be completed before 2030. This is due to the number of engineering manning resources required (over 100 man-years of engineering), and the need to spread these costs out over several years to fit the projects within the Navy's budget. And finally, due to the number of affected ships and the level of effort involved with these replacements, the installations would need to be done during ship maintenance availabilities. Due to the operating cycles of these ships, the maintenance availabilities for 20 ships would span approximately four (4) years. Since the installations cannot begin until all the engineering and logistics efforts are completed, the installations would not start until 2030, and not be completed until 2034. All these timelines are best case, with the assumption that the Navy budget can support these costs and all the efforts are completed on time and within budget.

Event	Time to complete
Planning, Estimating and Budgeting Process	2 Years
Engineering and Logistics Development	4 Years
Installation of Alternate Designs (20 ships)	4 Years
Total	10 Years

Table 4: Estimated Alternate Vendor System Development Schedule (Schedule Impact)

Since the Fairmount Automation legacy hardware is already obsolete, the Navy would not be able to keep these existing critical systems in operation long enough to complete this process. This would make these ships unusable for many years and would lead to unacceptable delays in fulfilling the agency's requirements (See 10 U.S.C. 3204(b)(B)(i) or 10 U.S.C. 3204(b)(B)(ii)).

6. Description of Efforts Made to Solicit Offers from as Many Offerors as Practicable.

In accordance with the Defense Federal Acquisition Regulation Supplement (DFARS) Procedures, Guidance, and Information (PGI) 206.302-1, a Request for Information (RFI) was posted to SAM.gov on 05 March 2024. Additionally, a synopsis of this procurement was published on the SAM.Gov on 20 March 2024. No contractors have responded to either posting as of the drafting of this J&A on 14 June 2024. No additional market research was conducted because it is not practicable for any other contractor than Fairmount Automation to provide the required supplies and services, for the reasons discussed in paragraph 5 above.

7. Determination of Fair and Reasonable Cost.

The Contracting Officer has determined that the anticipated cost to the Government for the supplies/services covered by the J&A will be fair and reasonable.

8. Actions to Remove Barriers to Future Competition.

For the reasons set forth in Paragraph 5, NSWCPD has no plans at this time to compete future contracts for the types of supplies/services covered by this document. Due to the limited remaining service life of the ship's involved, and the cost and time required for a redesign it is

not advantageous to consider alternative suppliers for these systems. If another design opportunity emerges for a different ship or different systems, NSWCPD will ensure a competitive solution is pursued.

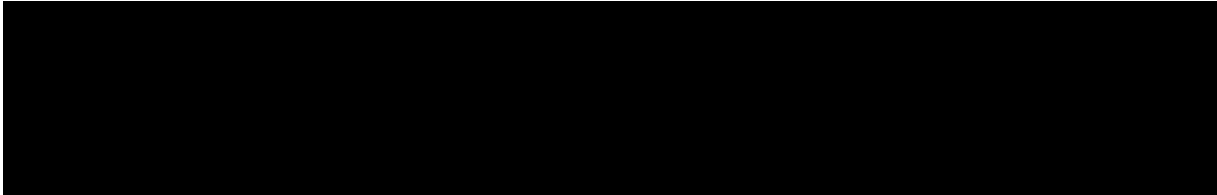
As this is a follow-on for supplies and services previously awarded on a non-competitive basis, in accordance with DFARS PGI 206.304(a)(S-70)(i), the prior J&A was provided to the approval authority for review. Approval of this J&A signifies the approval authority has made the determination required by DFARS PGI 206.304(a)(S-70)(ii).

CERTIFICATIONS AND APPROVAL

TECHNICAL AND REQUIREMENTS CERTIFICATION

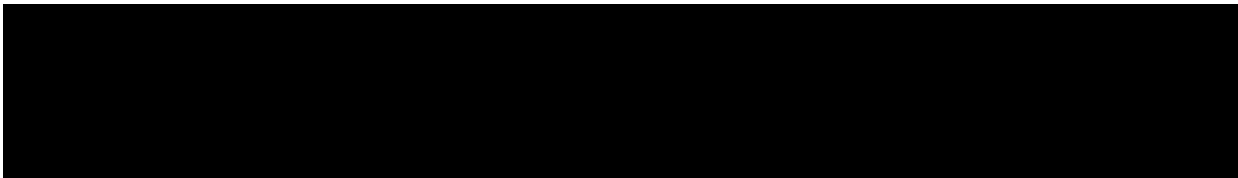
I certify that the facts and representations under my cognizance which are included in this Justification and its supporting acquisition planning documents except as noted herein are complete and accurate to the best of my knowledge and belief.

COMBINED TECHNICAL AND REQUIREMENTS COGNIZANCE:



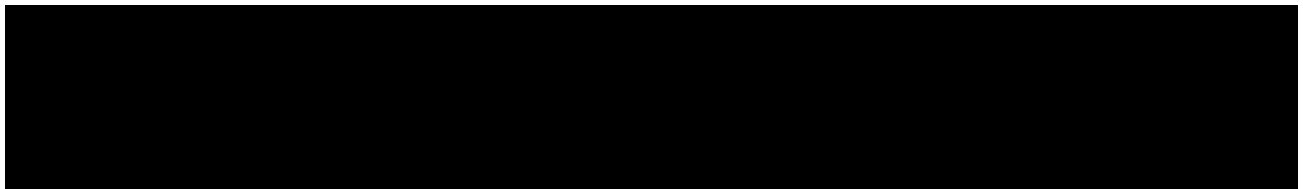
LEGAL SUFFICIENCY REVIEW

I have determined that this Justification is legally sufficient.



CONTRACTING OFFICER CERTIFICATION

I certify that this Justification is accurate and complete to the best of my knowledge and belief.



COMPETITION ADVOCATE APPROVAL

Upon the basis of the above Justification, I hereby approve, as Competition Advocate, the solicitation of the proposed procurement(s) described herein using other than full and open competition, pursuant to the authority of 10 U.S.C. 3204(a)(1).

